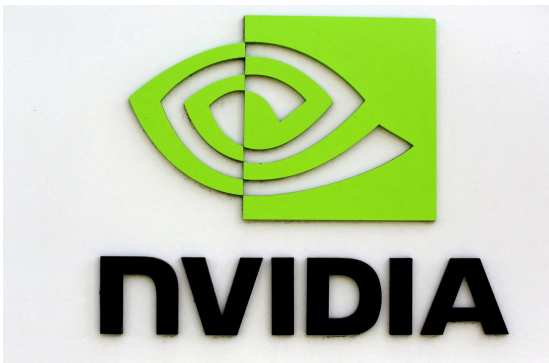




AI - Boom or Bust

ECE 143 Group 17

Katayoon Kaviani, Lakshya Goyal, Catherine Renteria, Ananya Kapoor, Ian DeGrood





Introduction

Comparing previous bubbles in the stock market (dot-com, EV, crypto) and comparing to the stocks trends of the AI dominated market

We want to determine whether we are experiencing an AI bubble by comparing current trends to trends seen in previous bubbles.

Which companies contributed most to the formation of the AI bubble?

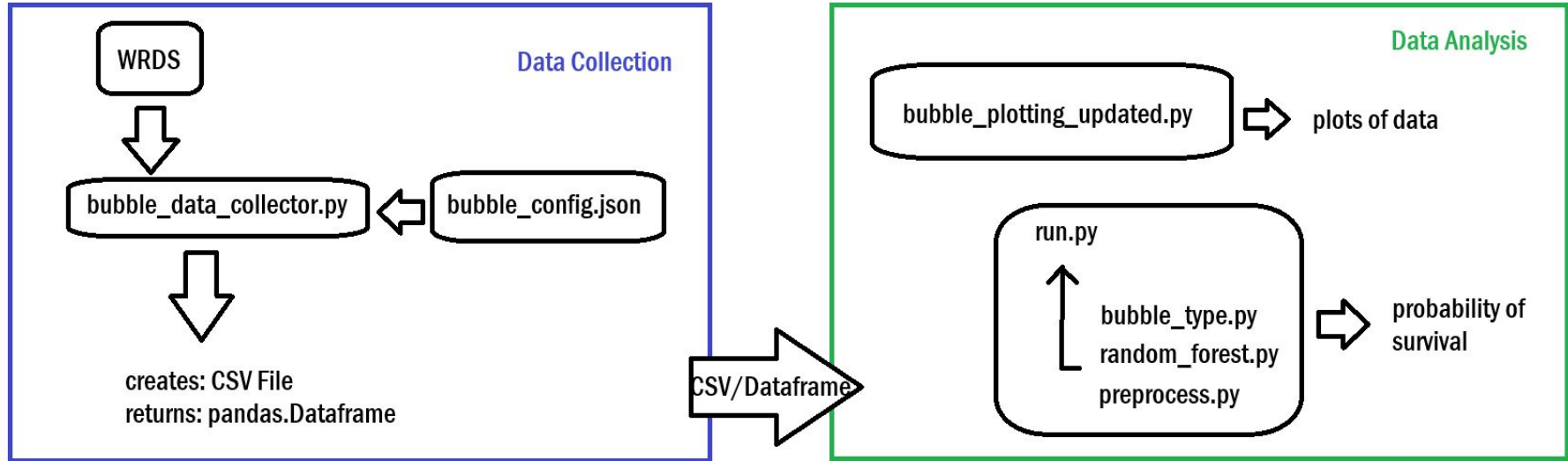
Determine which factors are important for company success post-bubble.

Make a prediction for how much companies will be affected by the bubble's pop



Data Overview

Using WRDS' (**Wharton Research Data Services**) database of CRSP (stocks) & Compustat (quarterly, annually stocks) merged together.



Metric Terms & How we derived them

Valuation Metrics

Metric	Formula	Meaning
Market Cap	$\text{Price} \times \text{Shares Outstanding}$	Total equity value of the firm
P/S (Price to Sales)	$\text{Market Cap} \div \text{Sales}$	How much investors pay per \$1 of revenue
P/E (Price to Earnings)	$\text{Market Cap} \div \text{Net Income}$	How much investors pay per \$1 of earnings
P/B (Price to Book)	$\text{Market Cap} \div \text{Book Equity}$	Compares market value to accounting book value
EV (Enterprise Value)	$\text{Market Cap} + \text{Total Liabilities} - \text{Book Equity}$	Value of entire firm (equity + debt)
EV/Sales	$\text{EV} \div \text{Sales}$	Revenue-based valuation including debt

Metric Terms & How we derived them

Profitability Metrics

Metric	Description
Profit margin	Net income generated per \$1 of sales
ROE	Net income as a percentage of equity
ROA	Net income as a percentage of total assets
Debt/Assets	Portion of assets financed by debt

Building the Bubble Index

1. For each firm, standardize key valuation ratios over time (for P/S, P/B, EV/Sales, P/E):

$$z_{i,t}(m) = \frac{m_{i,t} - \bar{m}_i}{\sigma_i(m)}$$

2. **Bubble Index (firm-day)** = average of the four z-scores

$$BI_{i,t} = \text{mean}\{z_{i,t}(\text{P/S}), z_{i,t}(\text{P/B}), z_{i,t}(\text{EV/S}), z_{i,t}(\text{P/E})\}$$

3. **Market-level Bubble Index:** aggregate across firms

Mean and max over firms for each date

$$BI_t^{\text{mean}} = \frac{1}{N_t} \sum_{i=1}^{N_t} BI_{i,t}$$

Dotcom Bubble:

...

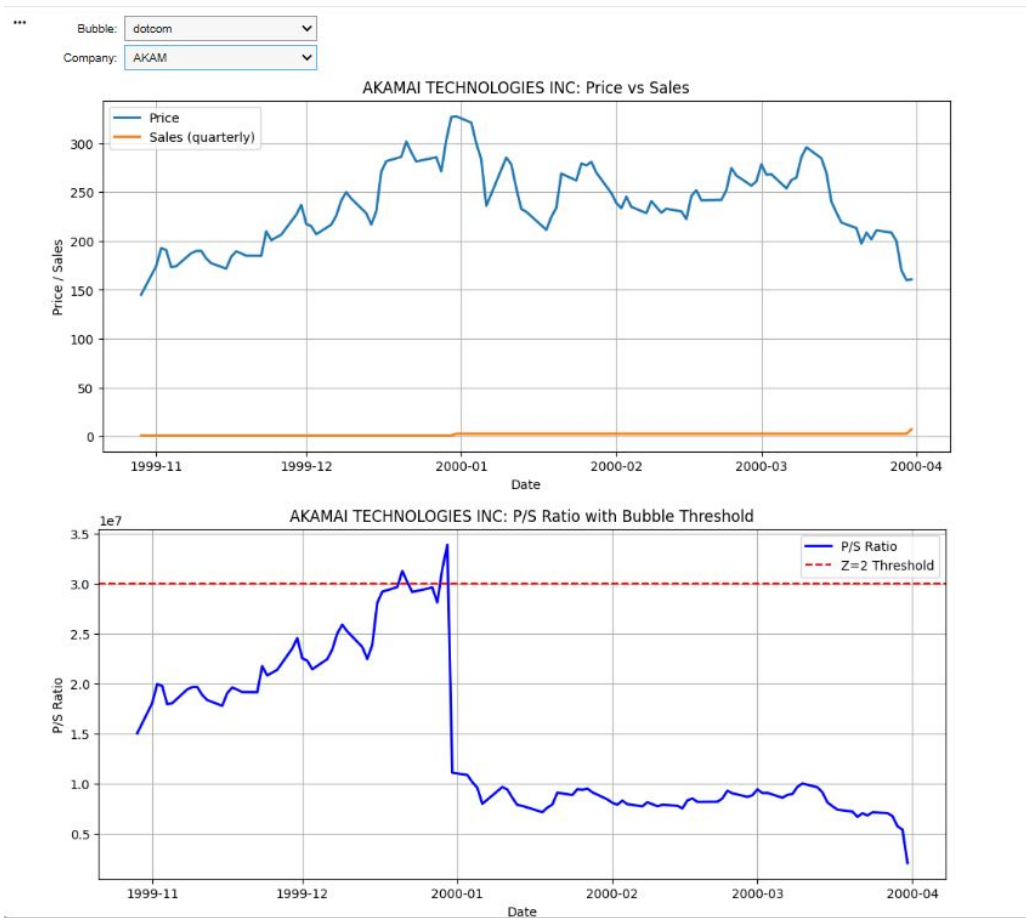
Bubble: dotcom ▼

Company: All companies ▼

Cumulative Bubble Index: DOTCOM

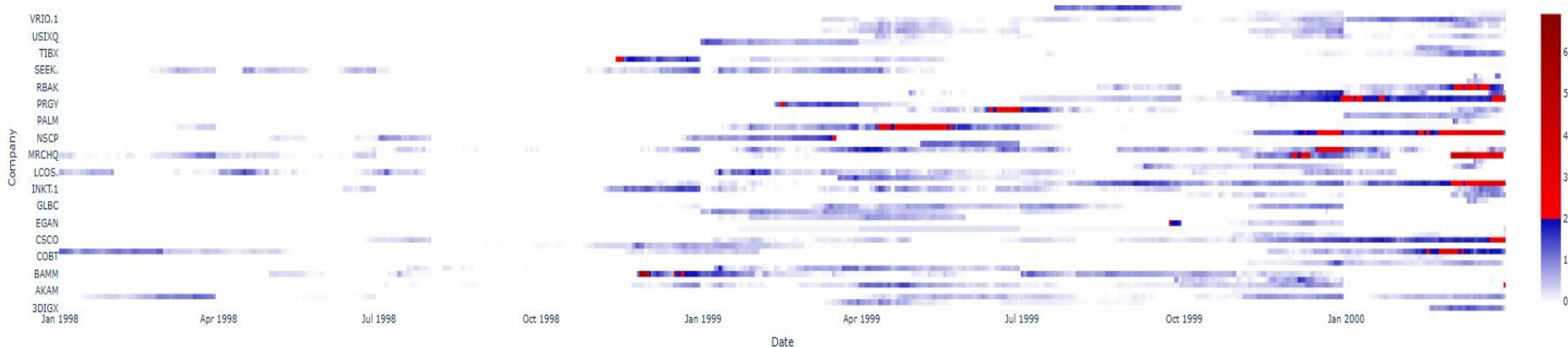


Dotcom Price-to-Sales



Dotcom Bubble Heatmap

dotcom: Bubble Index Heatmap



Results

EV Bubble:

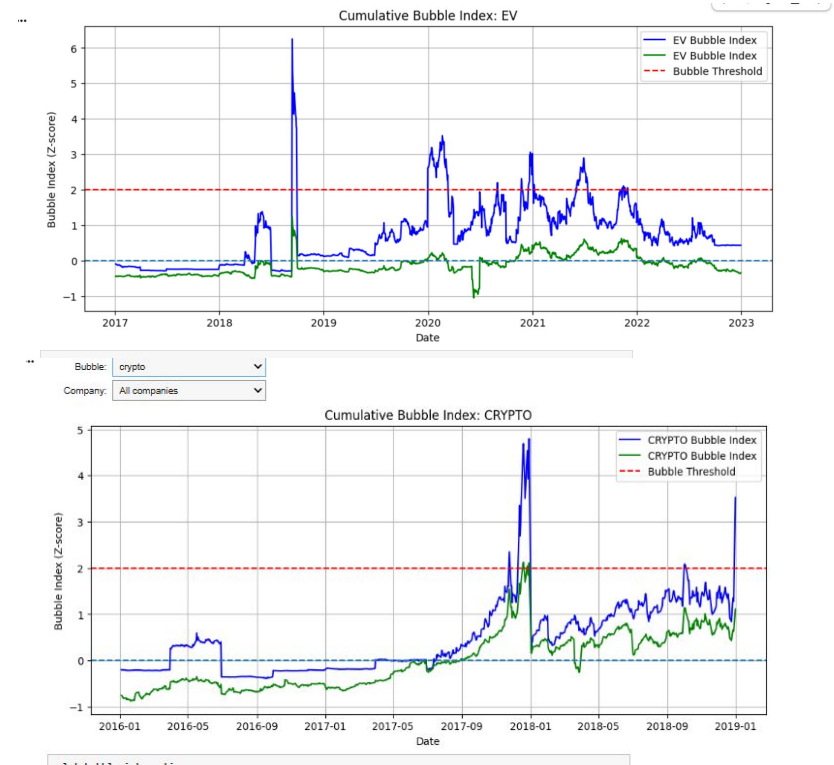
Peak between 2018-2019

Mean relatively low, peaks caused by outliers

Crypto Bubble:

Peaked in 2018

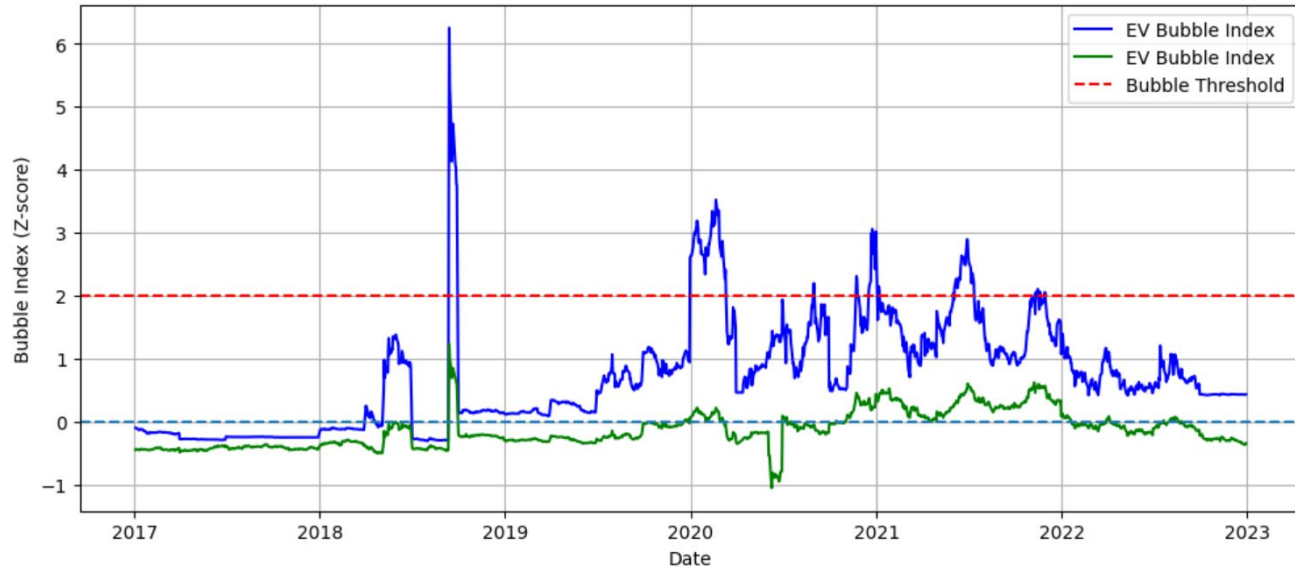
Mean peaked high, market as a whole overvalued



Results

EV Bubble:

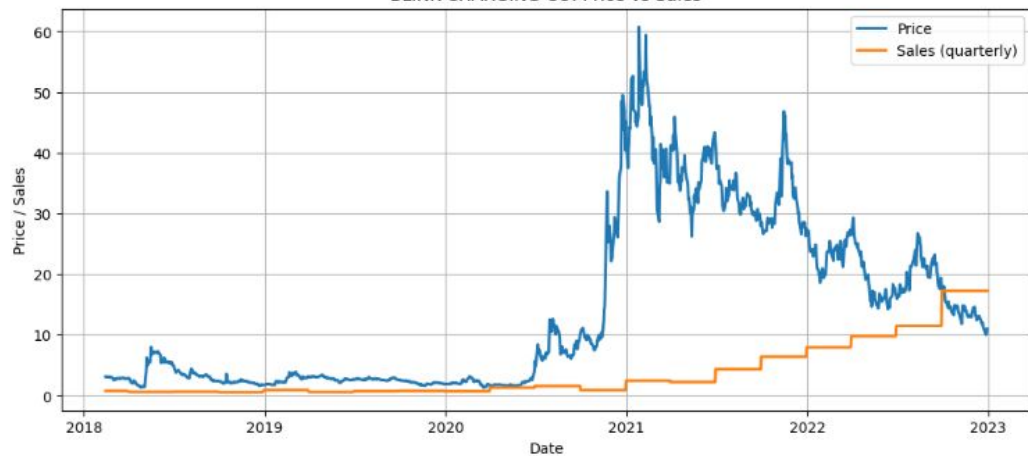
Cumulative Bubble Index: EV



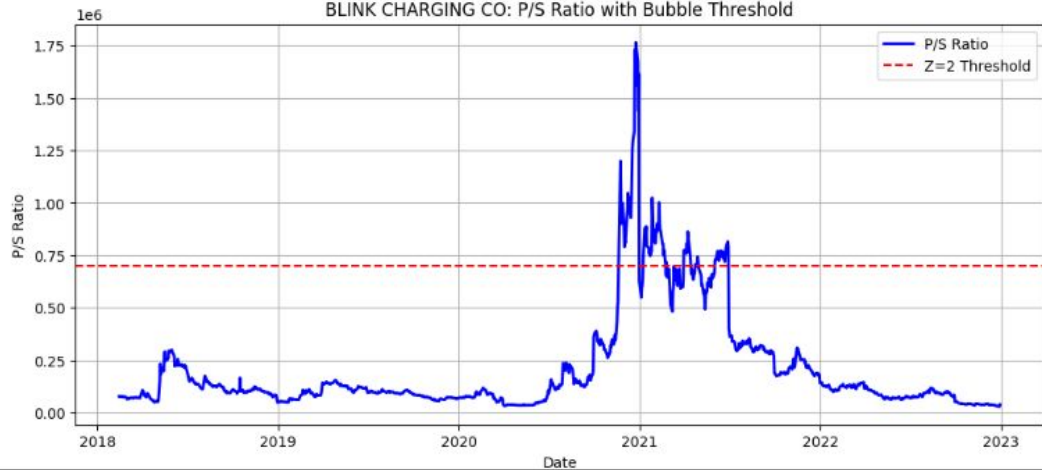
Bubble: ev ▼

Company: BLNK ▼

BLINK CHARGING CO: Price vs Sales

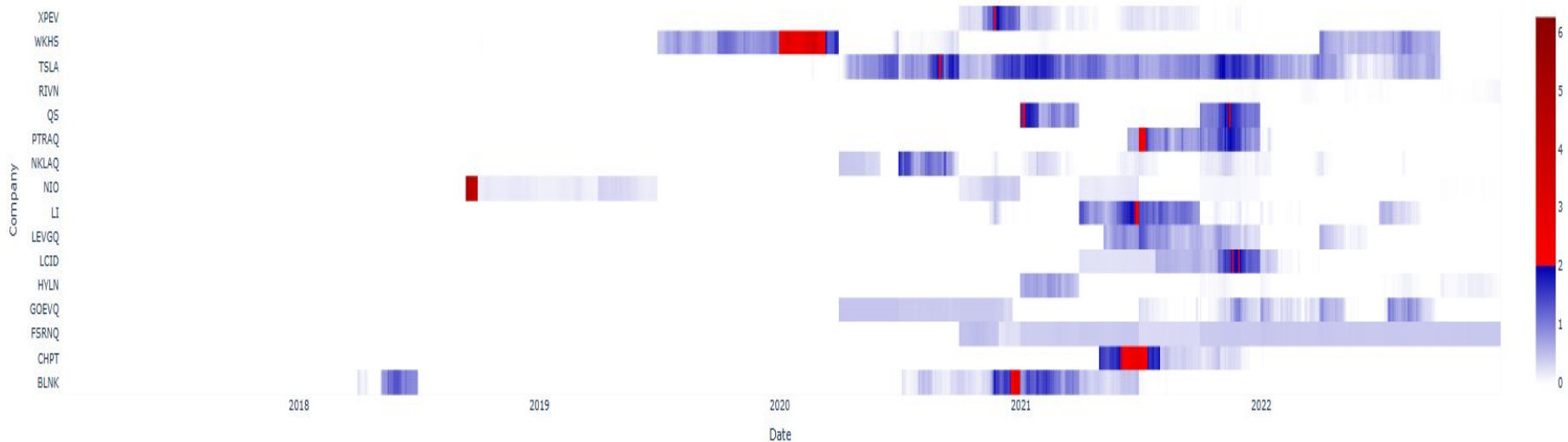


BLINK CHARGING CO: P/S Ratio with Bubble Threshold



EV Bubble Heatmap

ev: Bubble Index Heatmap

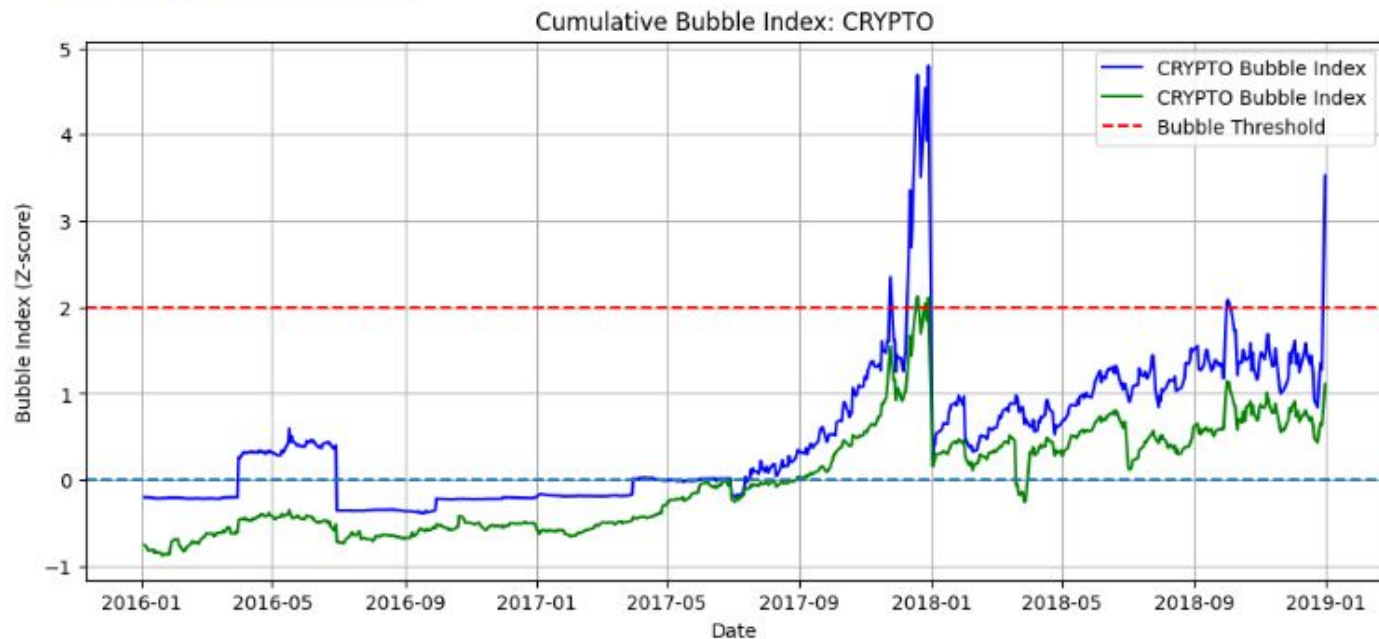


Crypto Bubble:

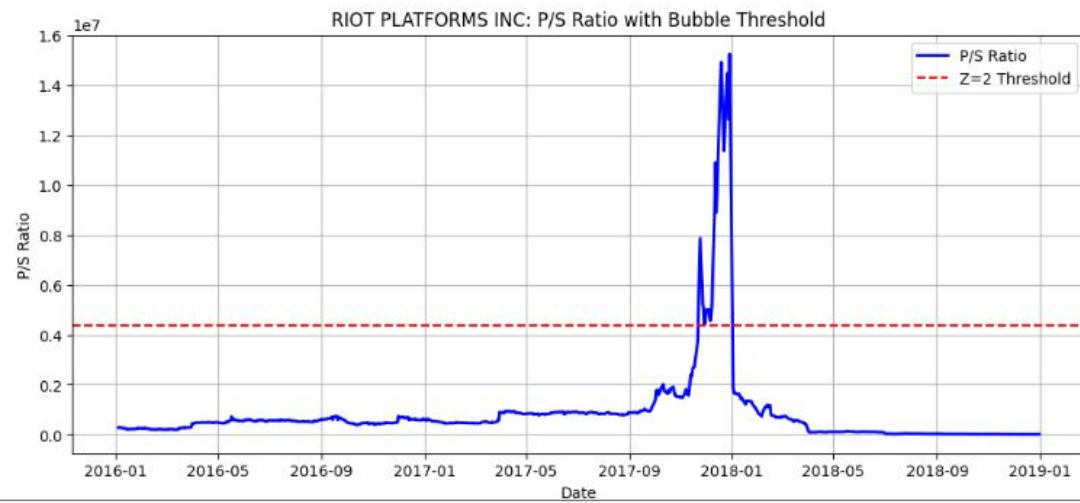
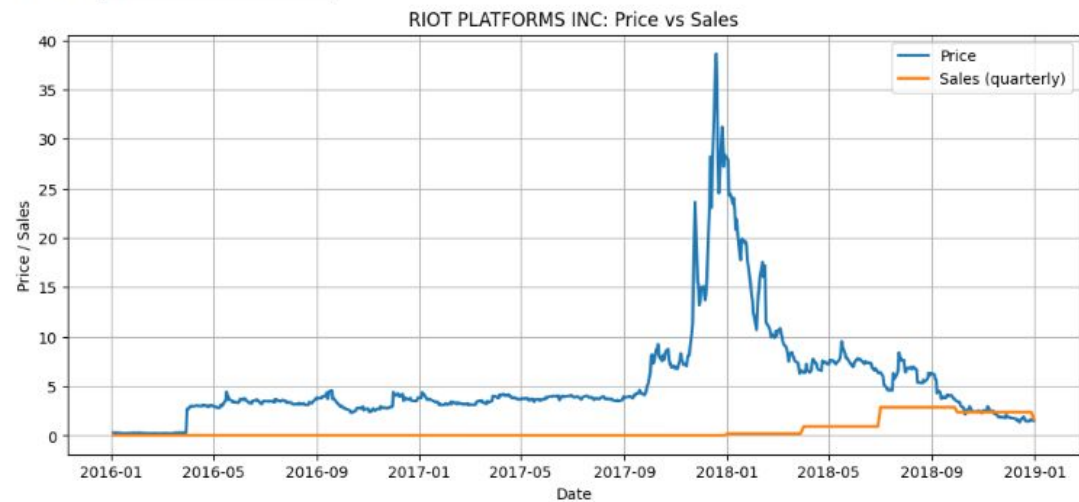
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Bubble: crypto

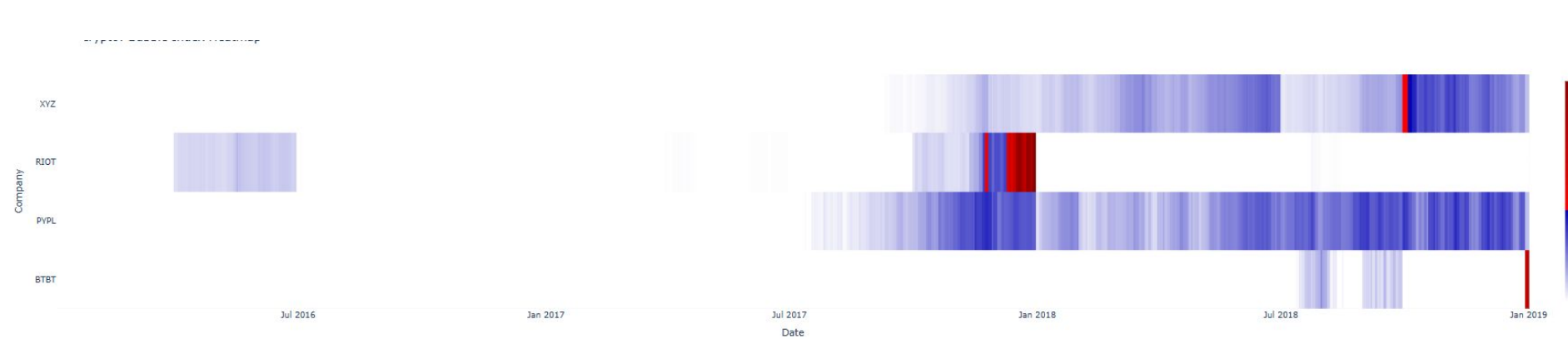
Company: All companies



Bubble: crypto
Company: RIOT



Crypto Bubble Heatmap



AI:

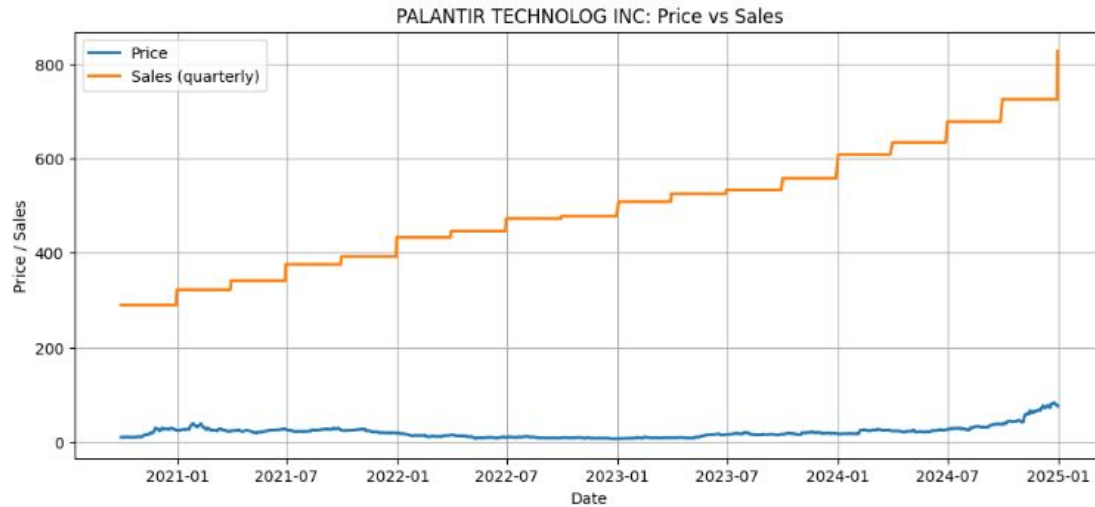
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Bubble: ai

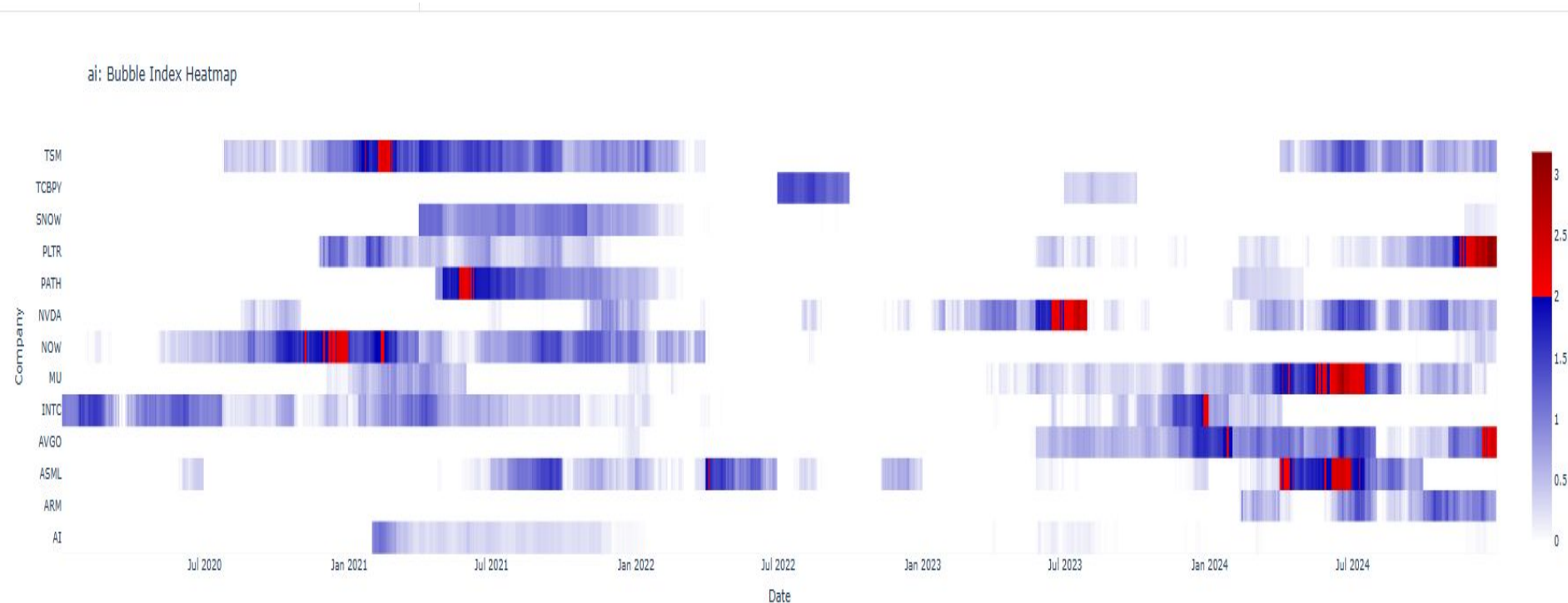
Company: All companies



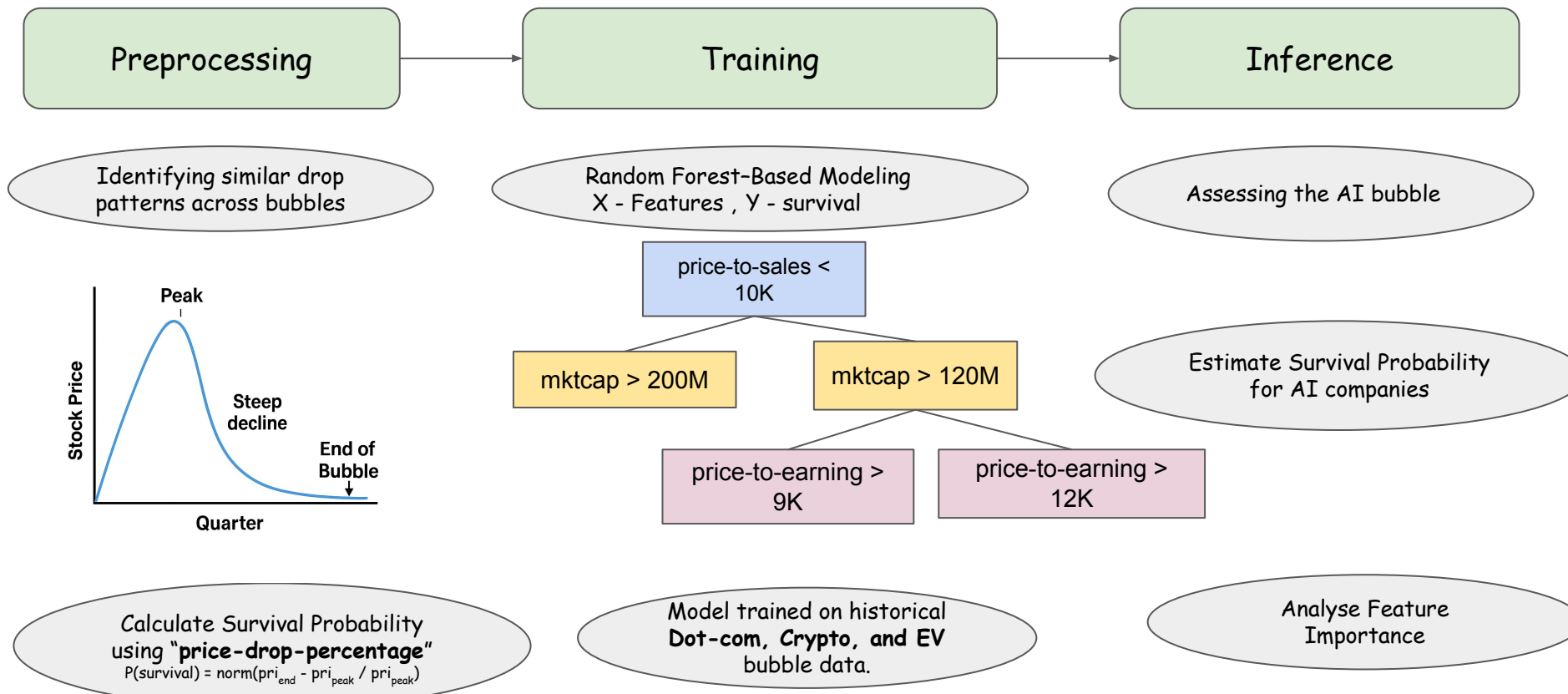
Bubble: ai
Company: PLTR



AI Heatmap



Survival Analysis : Modeling



Survival Analysis : Results

- When Price-to-Sale $\uparrow$, then Survival Rank $\downarrow$
- When Price-to-Book $\uparrow$, then Survival Rank $\downarrow$
- When Market-Cap $\uparrow$, then Survival Rank $\uparrow$

Company	Survival Rank	Market-Cap	Price-to-Book	Price-to-Earning
BROADCOM	1	319M	34K	128K
NVIDIA	2	949M	81K	283K
TAIWAN SEM.	3	113M	7K	181K
ARM HOLDINGS	7	13M	15K	766K
UIPATH INC	8	11M	48K	-148K

